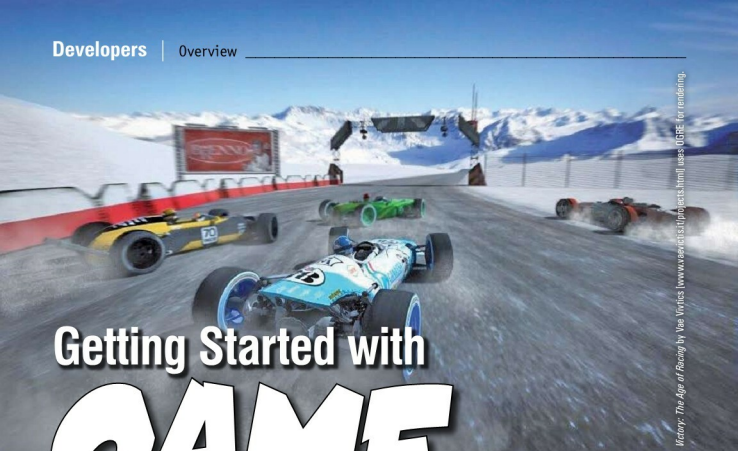




Getting Started with GAME Development

Victory: The Age of Racing by Vaa Virtues (www.virtues.com/projects.html) uses OCSE for rendering.

Creating a game is like creating a movie. It involves money, and the time and effort of a lot of people—or so I thought. Then I bumped into independent developers who were making games on shoestring budgets, yet keeping abreast with bigger production houses that were backed by million-dollar budgets.

Game development doesn't necessarily mean that you have to know programming, but this article focuses on development rather than anything else. Graphics programming is different from your regular application development. Regular applications are event-driven. Games are not.

Event-driven applications are almost all applications that you use every other day, like a calculator or a spreadsheet program—they require inputs from a user to move along.

When I started working with Visual Basic a long time ago, I didn't really know anything about how games worked. However, I was pretty used to the idea of event-driven applications.

Higher-level languages like Visual Basic take care of a lot of low-level work using intermediate runtimes. These leave you with a bit of reduced flexibility, unlike when dealing with the core platform itself.

Even later, after switching over to C++,

graphics programming still remained an elusive dream that left me confused because none of the applications I wrote behaved like games—they rarely 'acted' for that matter, and then came the game loop.

The game loop

The game loop is the heart of a game. It's a regular loop that continuously executes commands that keep the whole game world, as well as its occupants, running.

There were only ideas in the beginning. To move something on screen, I could use a loop or a keyboard input event—yes, even that was good. But my program flow was stuck in the loop of that specific event. If a car was being moved by my arrow keys, nothing else worked while that happened. To make my whole game world live and make everyone happy, the 'pros' of the black art of graphics programming came up with something called the game loop.

Regular applications waited around for